Continuous coupling trajectories — run report

RUN

194533_cross-sweep_short-low-power-verification
Status: completed | Started: 2026-09-29T18:45:33.565258+00:00

DIAGNOSTIC SUMMARY

Target actuator(s): phi1, phi2; sine: 0.1164 Hz; measured PAX rate: 6.88 Hz; saved RP reference rate: 381 Hz; estimated PAX samples/cycle: 59.13.
Recorded cycles: 4; bias values: phi1 target 1, phi2 target 1. Waveform tracking: deviations or missing/clipped reference present. Rising/falling branches (>5° descriptive threshold): strong separation present. Cycle repeatability (5° descriptive threshold): nonrepeatable above threshold. Bias-dependent plane changes (>5° descriptive threshold): unresolved. Low-DoP intervals: True. These thresholds are flags, not mechanism assignments.

PURPOSE

Continuously sine-drive phi1, phi2, or both while stepping and settling the orthogonal bias; record PAX plus measured OUT1/OUT2 references.

REQUIRED OPTICAL SETUP

PAX at final output; both paths open unless prompted. Wire OUT1 to IN1 and OUT2 to IN2 as voltage references. 'both' runs the complete diagnostic for both actuators automatically.

RECORDED BENCH CONTEXT (RECIPE)

In front of NPBS 1, C port
Red Pitaya (0-1 V command) -> Thorlabs MDT690 -> piezo

RUN COMMENT

Post-transition-fix short stress test: one low-power fixed bias per target, 1 warm-up + 4 recorded cycles, 2 s post-ramp settle. OUT2/IN2 reference known untrustworthy pending physical channel check.

OPERATOR SETUP NOTES

V_pi expectation only; verify channel and measured driver gain before calibration.

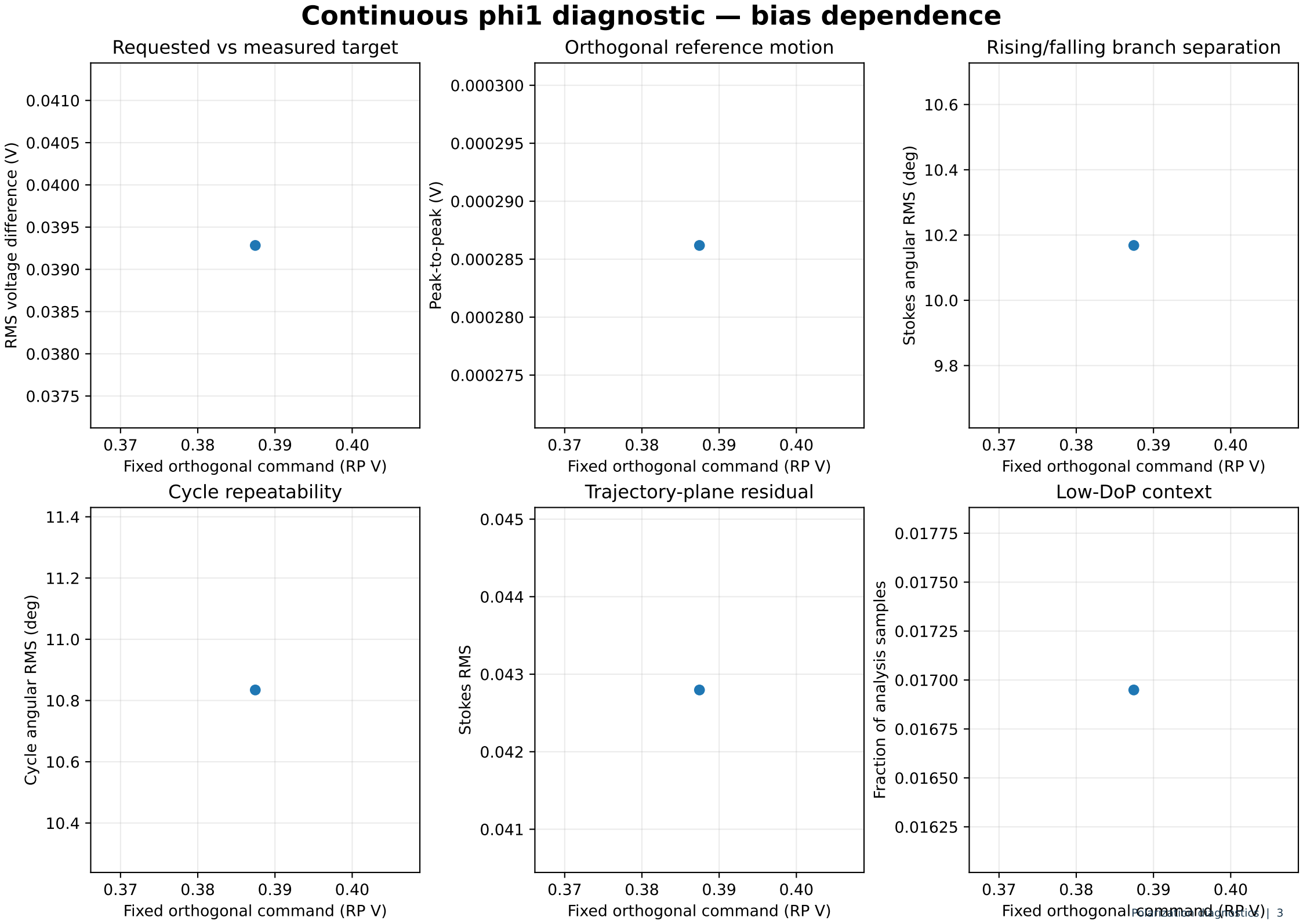
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DATA

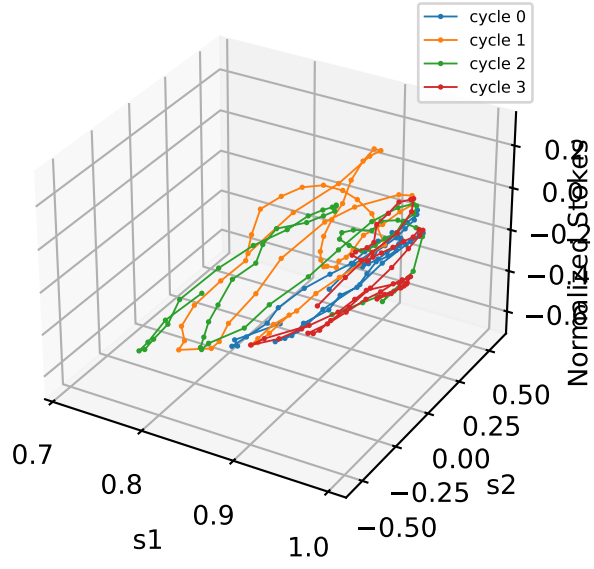
613 recorded samples; acquisition: 613
Source: data.csv; full acquisition settings: recipe.json; status: run.json

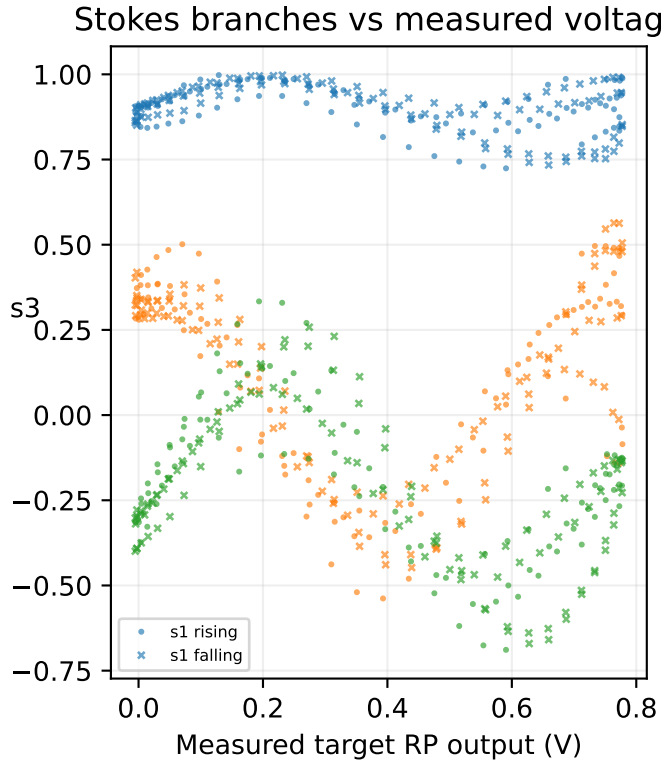
RAW DOP

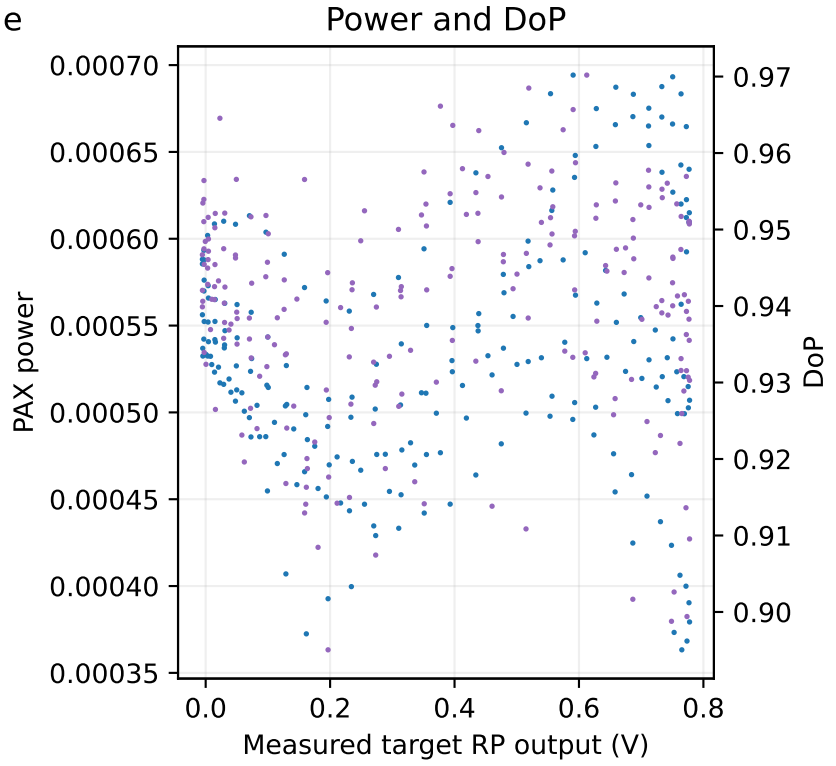
Mean 0.9308; range 0.7187 to 0.9702. 0 samples outside 0–1. Telemetry is not a new calibration.

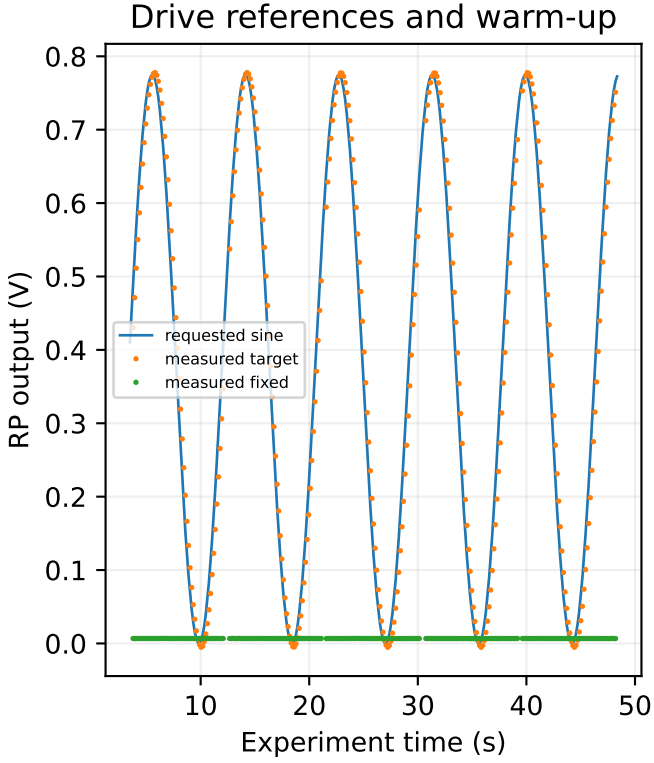

Continuous phi1 diagnostic — bias dependence
Requested vs measured target
RMS voltage difference (V)
0.0410
0.0405
0.0400
0.0395
0.0390
0.0385
0.0380
0.0375
0.37
0.38
0.39
0.40
Fixed orthogonal command (RP V)
Orthogonal reference motion
Peak-to-peak (V)
0.000300
0.000295
0.000290
0.000285
0.000280
0.000275
0.37
0.38
0.39
0.40
Fixed orthogonal command (RP V)
Rising/falling branch separation
Stokes angular RMS (deg)
10.6
10.4
10.2
10.0
9.8
0.37
0.38
0.39
0.40
Fixed orthogonal command (RP V)
Cycle repeatability
Cycle angular RMS (deg)
11.4
11.2
11.0
10.8
10.6
10.4
0.37
0.38
0.39
0.40
Fixed orthogonal command (RP V)
Trajectory-plane residual
Stokes RMS
0.045
0.044
0.043
0.042
0.041
0.37
0.38
0.39
0.40
Fixed orthogonal command (RP V)
Low-DoP context
Fraction of analysis samples
0.01775
0.01750
0.01725
0.01700
0.01675
0.01650
0.01625
0.37
0.38
0.39
0.40
Fixed orthogonal command (RP V)

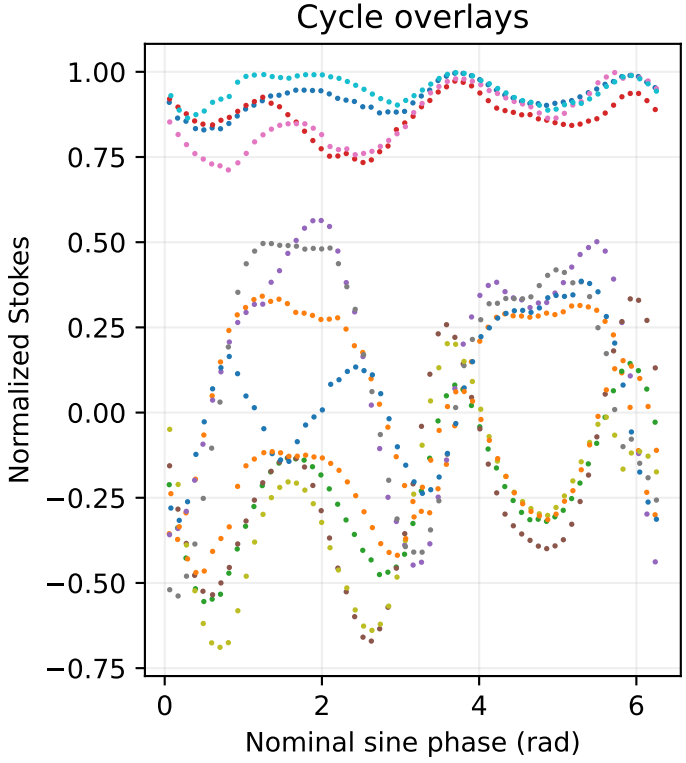
phi1 sine; fixed phi2 = 0.3875 RP V (segment 0)



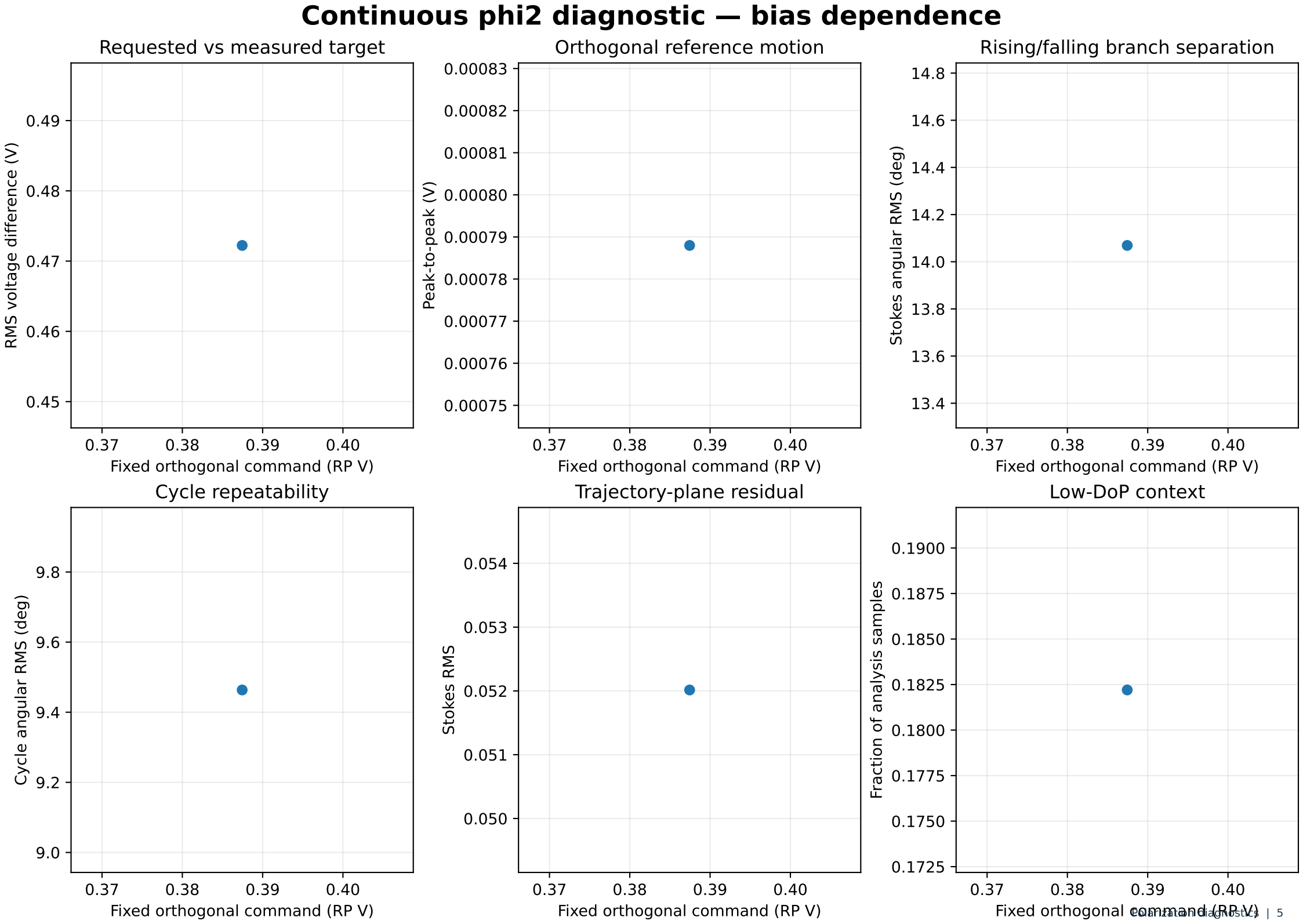




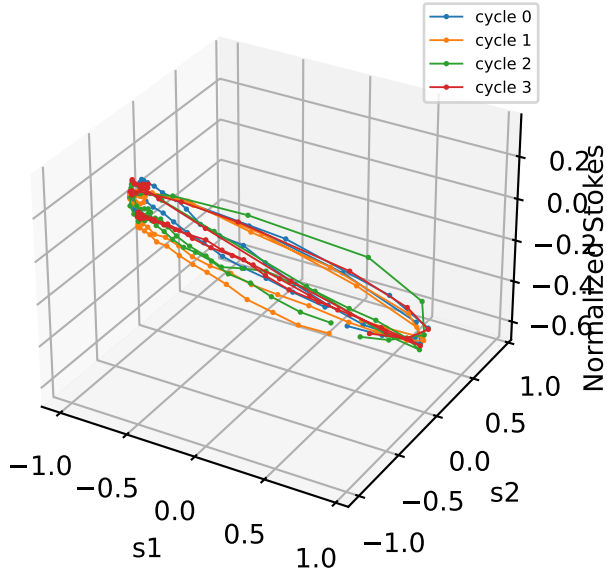




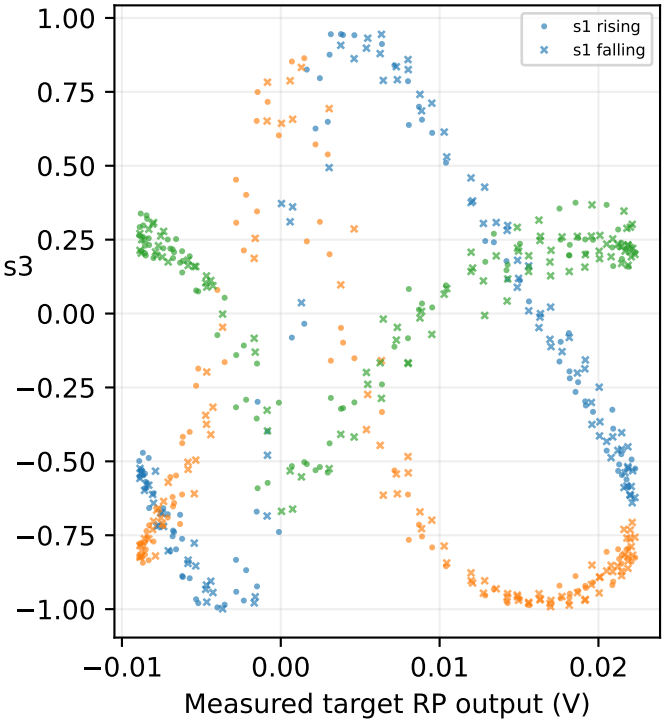
Ideal reference: phi1 should keep s1 fixed and total power independent of drive.
Plane normal: [0.970, 0.025, -0.241]
Plane residual RMS: 0.0428
Rising/falling angular RMS: 10.2°
Rising/falling power RMS: 5.74e-05 PAX units
Cycle angular RMS: 10.8°
Cycle power RMS: 7.99e-05 PAX units
Cycle DoP RMS: 0.0148
Tracking RMS: 0.03928 V
Reference coverage: 95.8%
Branch separation is reported descriptively; it is not assigned to piezo hysteresis.
Cycle 0 plane: [0.98, -0.02, -0.22], residual 0.0116
Cycle 1 plane: [0.99, 0.02, -0.17], residual 0.0373
Cycle 2 plane: [0.95, 0.01, -0.32], residual 0.0346
Cycle 3 plane: [0.98, 0.04, -0.18], residual 0.0223


Continuous phi2 diagnostic — bias dependence
Requested vs measured target
RMS voltage difference (V)
0.49
0.48
0.47
0.46
0.45
0.37
0.38
0.39
0.40
Fixed orthogonal command (RP V)
Orthogonal reference motion
Peak-to-peak (V)
0.00083
0.00082
0.00081
0.00080
0.00079
0.00078
0.00077
0.00076
0.00075
0.37
0.38
0.39
0.40
Fixed orthogonal command (RP V)
Rising/falling branch separation
Stokes angular RMS (deg)
14.8
14.6
14.4
14.2
14.0
13.8
13.6
13.4
0.37
0.38
0.39
0.40
Fixed orthogonal command (RP V)
Cycle repeatability
Cycle angular RMS (deg)
9.8
9.6
9.4
9.2
9.0
0.37
0.38
0.39
0.40
Fixed orthogonal command (RP V)
Trajectory-plane residual
Stokes RMS
0.054
0.053
0.052
0.051
0.050
0.37
0.38
0.39
0.40
Fixed orthogonal command (RP V)
Low-DoP context
Fraction of analysis samples
0.1900
0.1875
0.1850
0.1825
0.1800
0.1775
0.1750
0.1725
0.37
0.38
0.39
0.40
Fixed orthogonal command (RP V)

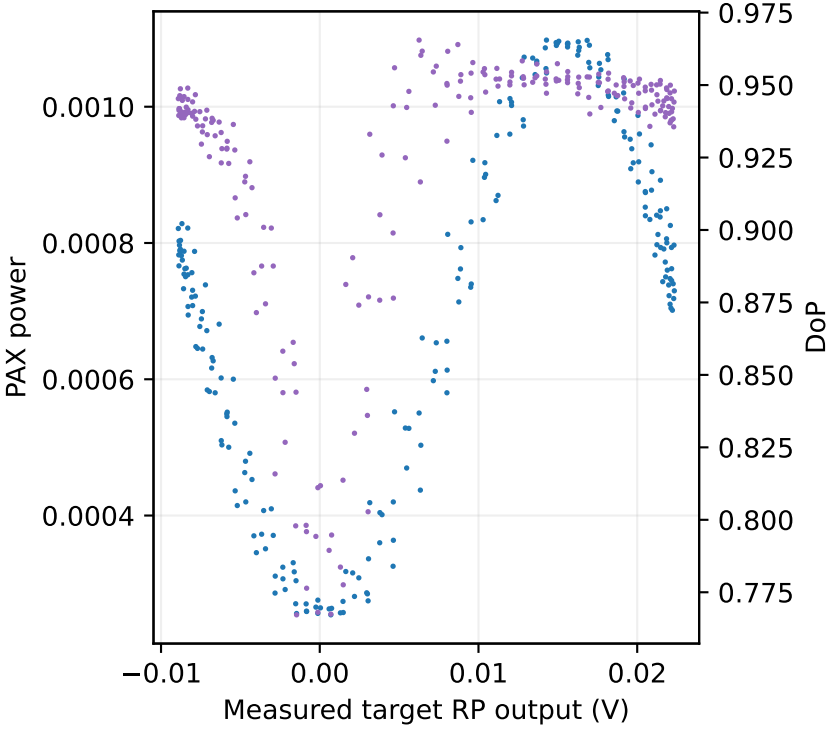
phi2 sine; fixed phi1 = 0.3875 RP V (segment 1)



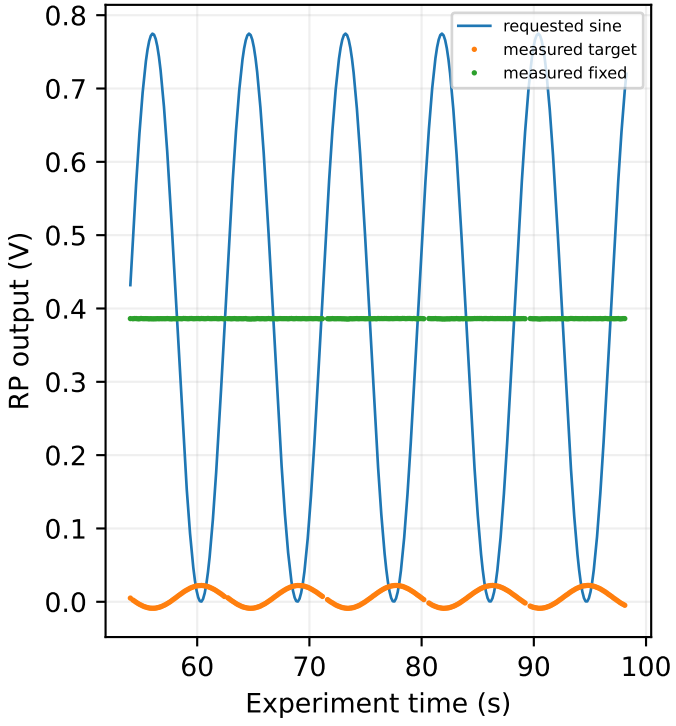
Stokes branches vs measured voltage



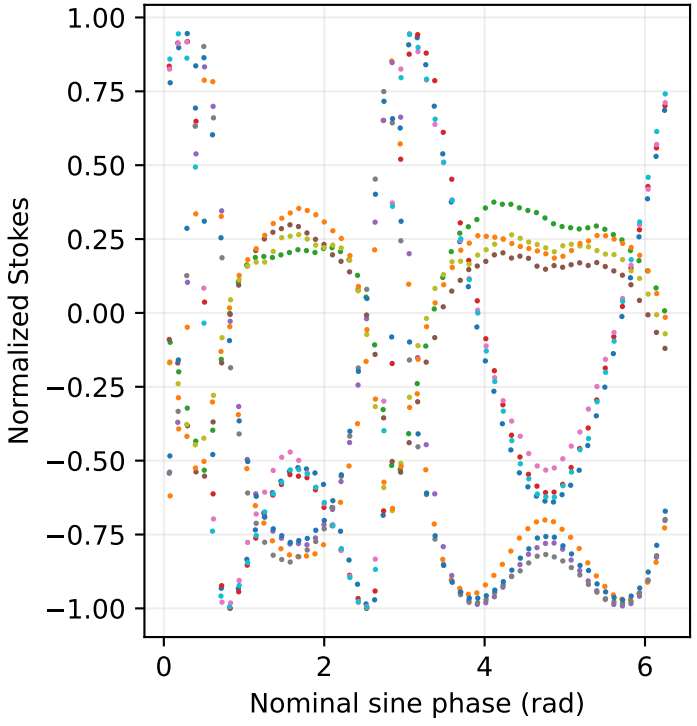
Power and DoP



Drive references and warm-up



Cycle overlays

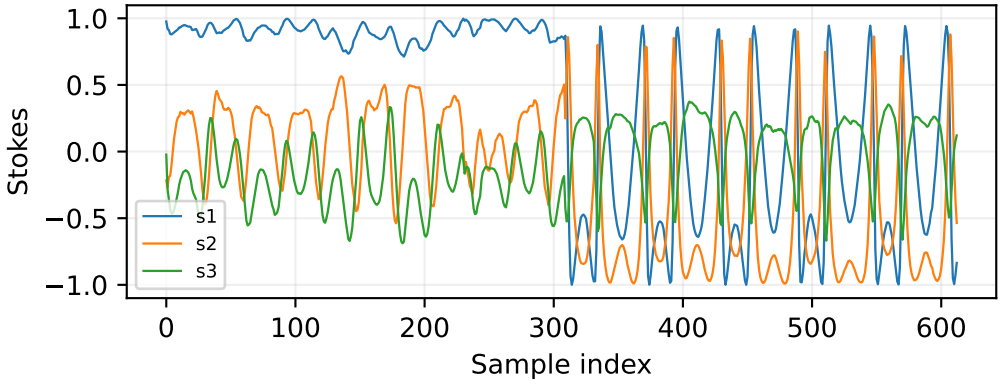


Ideal reference: phi2 changes polar coordinate; fixed phi1 sets azimuth orientation.
Plane normal: [0.163, 0.385, 0.908]
Plane residual RMS: 0.05201
Rising/falling angular RMS: 14.1°
Rising/falling power RMS: 5.86e-05 PAX units
Cycle angular RMS: 9.46°
Cycle power RMS: 4.19e-05 PAX units
Cycle DoP RMS: 0.0129
Tracking RMS: 0.4722 V
Reference coverage: 97.0%
Branch separation is reported descriptively; it is not assigned to piezo hysteresis.
Cycle 0 plane: [0.16, 0.40, 0.90], residual 0.0477
Cycle 1 plane: [0.17, 0.35, 0.92], residual 0.0401
Cycle 2 plane: [0.16, 0.38, 0.91], residual 0.0459
Cycle 3 plane: [0.16, 0.42, 0.90], residual 0.0441

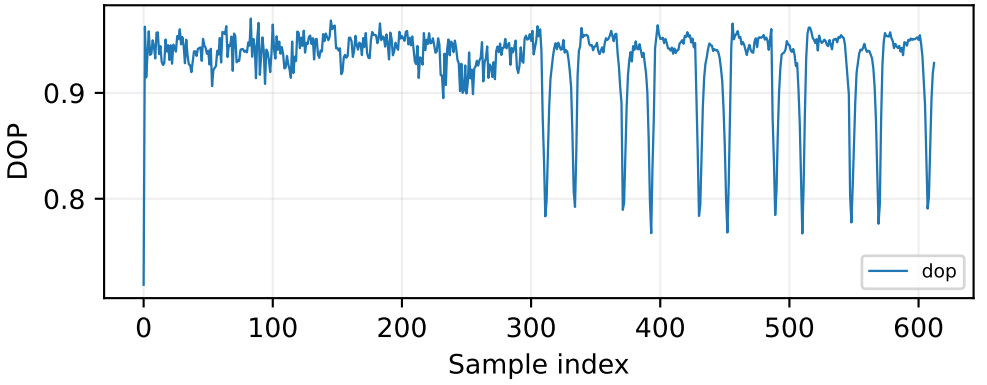
Recorded telemetry

Acquisition order preserves stage changes; no timing alignment or transfer gain is inferred.

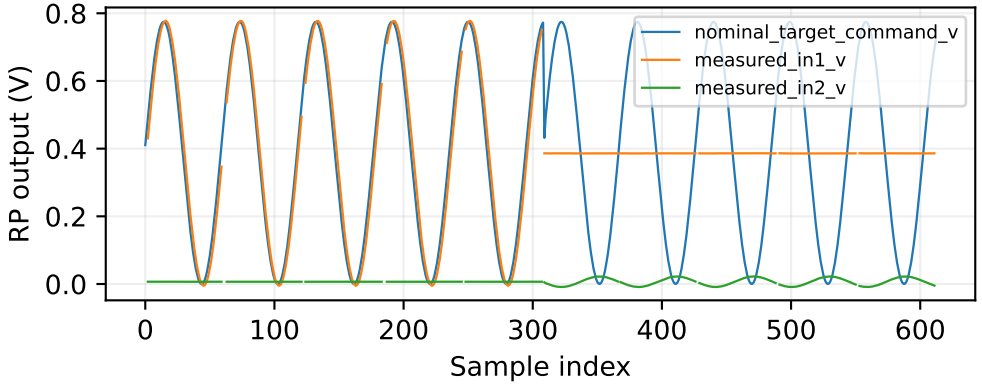
Normalized Stokes



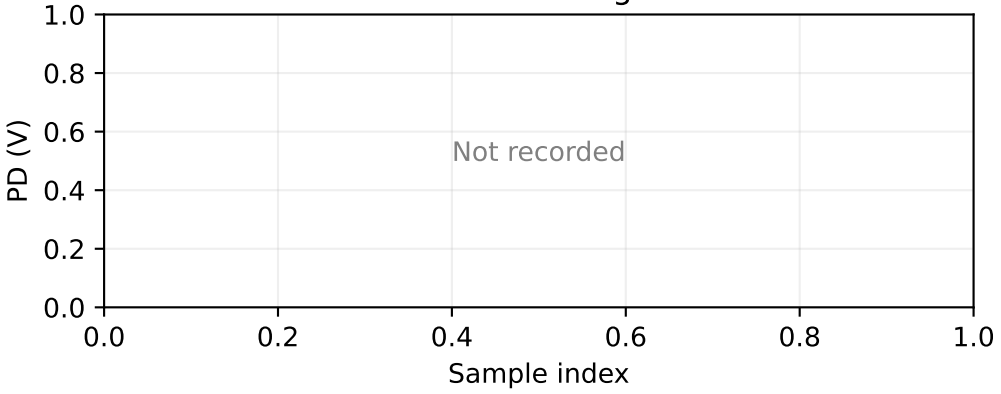
Raw degree of polarization



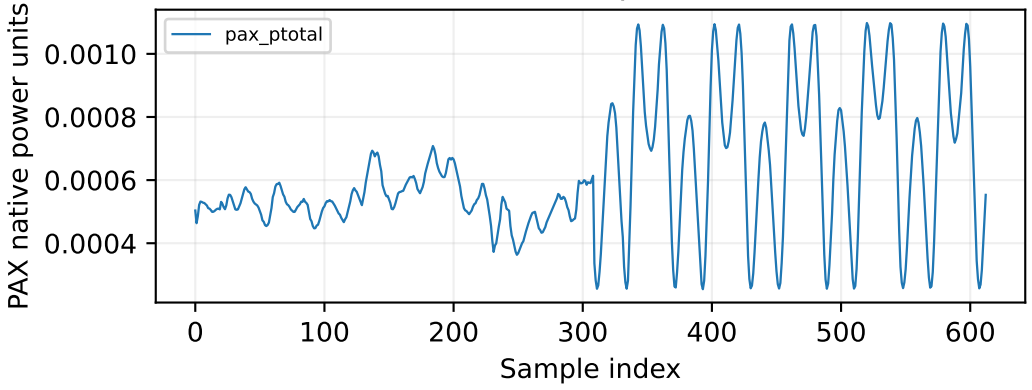
Actuator command / reference history



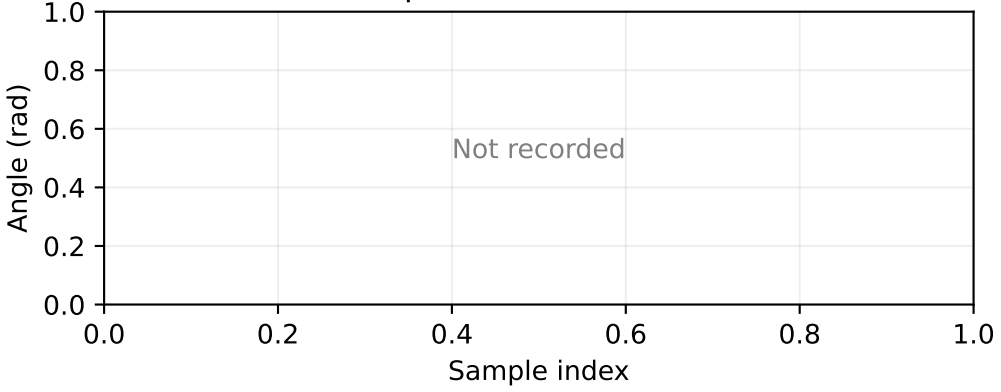
Photodiode signal



PAX total power



Sphere coordinates



Active parameters

RP commands: volts, bounded to 0–1. V_lambda: terminal volts for 2π. Gains: terminal V / RP V.

Parameter	Recorded value
label	"short-low-power-verification"
comment	"Post-transition-fix short stress test: one low-power fixed bias pe
actuator	"both"
report	"pdf"
pax_rotation_velocity_hz	60.0
pax_measurement_mode	5
pax_wavelength_nm	830.0
phi1_v_lambda	13.078125
phi2_v_lambda	13.078125
phi1_actuator_volts_per_rp_volt	16.875
phi2_actuator_volts_per_rp_volt	16.875
rp_output_min_voltage	0.0
rp_output_max_voltage	1.0
cross_sweep_sine_frequency_hz	0.1
cross_sweep_sine_center_voltage	0.3875
cross_sweep_sine_amplitude_voltage	0.3875
cross_sweep_warmup_cycles	1
cross_sweep_recorded_cycles	4
cross_sweep_phi1_bias_voltages	[0.3875]
cross_sweep_phi2_bias_voltages	[0.3875]
cross_sweep_settle_s	2.0
cross_sweep_bias_ramp_s	1.0
cross_sweep_bias_ramp_updates_per_s	50.0
cross_sweep_reference_sample_rate_hz	400.0

Active parameters

RP commands: volts, bounded to 0–1. V_lambda: terminal volts for 2π. Gains: terminal V / RP V.

Parameter	Recorded value
cross_sweep_scope_block_s	8.0
cross_sweep_min_pax_samples_per_cycle	20.0